

JAPAN AUTOMOBILES

Humanoids & Auto: Building Physical AI Together

We are significantly raising our demand forecasts for the humanoid robot market. Continuous advancements in physical AI are accelerating the development of humanoid robots, with an increasing number of players transitioning toward larger-scale prototypes and commercial production. Accordingly, we raise our 2035 unit sales forecast to 6.48 mn units from 1.38 mn previously. We now project the market size will reach \$138 billion (approximately ¥22 tn) by 2035, compared with our previous estimate of around ¥6 tn.

- While Japanese companies have secured an estimated 30–40% global market share in the FA (Factory Automation) sector, they lag significantly in the humanoid sector, where US, Chinese, and South Korean companies are actively working to establish early mass production systems. Nevertheless, Japan has a strong foundation in robotics, ranking fourth globally in industrial robot density and second only to South Korea in receptiveness to robotics adoption. Furthermore, Japan's well-developed automotive and auto parts supply chain provides a solid foundation for the mass production of actuators and other critical robot hardware components.
- We believe the automotive industry, which mass-produces one of the world's most complex industrial products at scale, is uniquely positioned to support the mass production of humanoid robots. We have scored global OEMs across three core dimensions: **Will** (intent to enter the humanoid market), **Should** (stagnation of core business), and **Could** (financial and operational capability). Currently, Tesla and Hyundai are among the most active participants in humanoid robotics. Among Japanese automakers, Toyota, Honda, and Mitsubishi Motors are relatively well positioned.
- We believe the Toyota Group, in particular, possesses all four elements required to establish a meaningful presence in robotics: **Needs** (securing labor), **Urgency** (addressing supply chain pressures associated with the shift to BEVs), **Technology** (hardware/software and mass production), and **Commitment** (management's positioning of robotics as a growth strategy, supported by financial strength). Assuming continued technological progress and successful mass production ramp-up, Toyota could potentially produce approximately 190,000 to 540,000 humanoid robot units in 2035, implying a global market

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share of approximately 3-8%. We estimate the incremental upside to our 3/2027 gross profit at 2-6%.

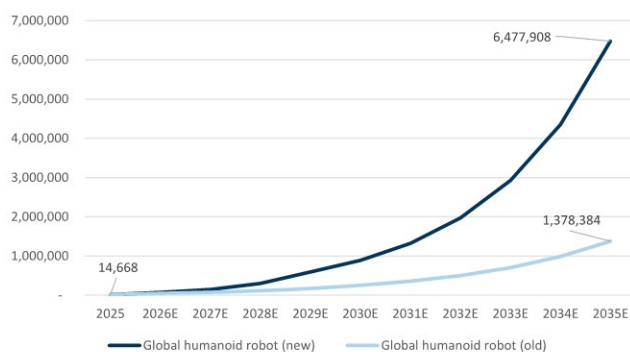
- Potential stock beneficiaries include **Toyota, Honda, Mitsubishi Motors, JTEKT** (initiating coverage with a Buy rating), **Aisin, MinebeaMitsumi, Renesas, Harmonic Drive**, and **NEC/Fujitsu** (through Noetra).
- Key risks to our view include: (1) overly optimistic expectations regarding the value of on-site data and master craftsmanship, (2) reliance on overseas Foundation models, and (3) the inability to leverage strong successful experiences in FA.

PM summary

Humanoids have significant growth potential

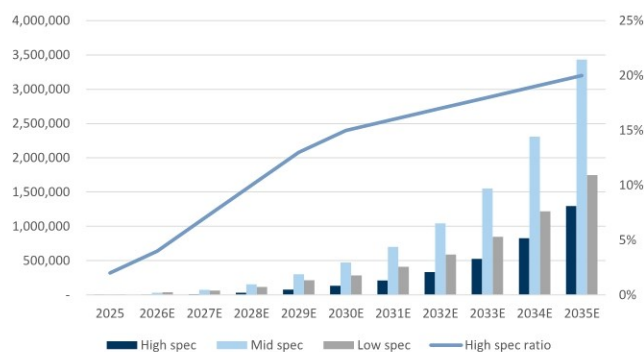
We are significantly raising our demand forecasts for the humanoid robot market. We raise our 2035 unit sales forecast to 6.48 mn units from 1.38 mn units previously. We project the market size will reach \$138 bn (approximately ¥22 tn) in 2035, compared to our previous estimate of around ¥6 tn. This upward revision is driven by the continuous improvement of physical AI technology, with humanoid robots from several makers increasingly transitioning toward larger-scale prototypes and commercial production. Furthermore, the industry is benefiting significantly from new open-source tools, rising funding, and government support. In addition, increasing deployments are expected to generate a positive flywheel effect, where robots in operation help gather real-world data to further train and improve AI software. Consequently, we believe that robotics, including humanoids, will become more general-purpose in utility over time.

Exhibit 1: High growth potential but also high uncertainty
Humanoid robot market outlook



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 2: Higher spec
Humanoid robot spec mix



Source: Company data, Goldman Sachs Global Investment Research

Close connection between robotics and the automotive industry

Automobile production bases, which possess a certain level of financial strength and have a pressing need for automation, will likely be early adopters. We think the data accumulated in the production bases and the cost reductions driven by mass production effects will serve as a catalyst for the spread of robots to other industries. Labor shortages and wage increases are particularly urgent issues for the automotive industry, making the utilization of physical AI indispensable. Furthermore, the automotive industry is facing an era where the shift to BEVs is reducing the number of parts from an average of about 30,000 to around 10,000 to 20,000. Actuators and precision reduction gears used in robot joints have a high affinity with automotive powertrain manufacturing, and we believe that many auto parts suppliers are eager to pivot their businesses and enter the robotics industry.

Auto industry is Well-Positioned to Pivot to Humanoid Robotics

We believe that the automotive industry—which mass-produces automobiles, one of the world's most complex industrial products, by the millions—is uniquely suited for the mass production of humanoid robots. We have scored global OEMs based on three core dimensions: Will (intent to enter the humanoid market), Should (stagnation of core

business), and Could (financial and operational capability). Currently, Tesla and Hyundai are the most actively engaged in humanoid robotics. Among Japanese automakers, Toyota, Honda, and Mitsubishi Motors are relatively well-positioned.

Will: Companies with concrete mass-production plans for humanoid robots are scored “2”, those with an intent or plan to enter are scored “1”, and others are scored “0”. Investments in humanoid-related companies were also factored into this score.

Should: Companies losing market share in their core automotive business are scored “2”, while those maintaining or growing their share are scored “1”. The trend of operating profit margins over the past three years was also taken into account.

Could: Companies with an annual global production volume of 3 million units or more are scored “2”, those with 1 million to 3 million units are scored “1”, and those with under 1 million units are scored “0”. The absolute amount of net cash in the automotive business was also factored in.

Exhibit 3: Will/Should/Could analysis

Humanoid robot readiness

	Will		Should		Could		Total score
	Expressed Interest	Minority/Majority Stakes	Core TAM Pressure	Margin pressure	Global Scale & Right Markets	Financial Ability	
Tesla		2	2	1	1	2	9
Hyundai		2	2	1	1	2	9
Toyota		1	2	1	1	2	8
BMW		1	1	2	2	1	6
Mercedes		1	1	2	2	1	6
Honda		1	1	2	2	1	6
Mitsubishi		1	1	2	2	0	5
VW		0	0	2	2	2	5
GM		0	0	2	2	2	5
Ford		0	0	2	2	2	5
Nissan		0	0	2	2	1	3
Suzuki		0	0	1	0	2	3
Mazda		0	0	2	2	1	3
SUBARU		0	0	2	2	0	2

Source: Company data, Goldman Sachs Global Investment Research

Low presence of Japanese manufacturers

While Japanese companies have secured a 30-40% global market share in the FA (Factory automation) sector, they are lagging significantly in the humanoid sector, where US, Chinese, and South Korean companies are actively working to establish early mass production systems. In fact, there is an increasing number of examples overseas where automobile/FA manufacturers are establishing joint ventures with humanoid companies. Furthermore, in South Korea, traditional parts manufacturers have actually started shipping hardware such as actuators for humanoid robots. Chinese manufacturers account for the majority of humanoid shipments in 2025. However, we believe that there are still opportunities for late-coming manufacturers as expanding the scope of applications requires further innovation in AI capabilities and robot hands.

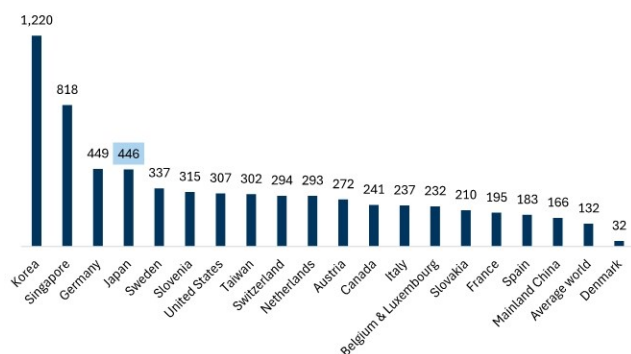
Sorting out Japan's position

Japan's physical AI sector is still in its infancy, but we think there are potential opportunities. The robotics industry has taken root in Japan, primarily in FA (such as precision reduction gears and planetary rollers), with the density of industrial robots ranking fourth in the world and the receptiveness to the robotics industry ranking second only to South Korea. Furthermore, the robust supply chain for automobiles and auto parts provides the groundwork for achieving mass production of actuators and

other components essential for robot hardware. The Japanese government has also positioned physical AI as a strategic sector and set a target of 10 million AI robots by 2040. Anticipating roles both as a “user” to support a shrinking workforce and as a “creator” to generate new industries (as a countermeasure against the hollowing out caused by the shift to BEVs), a support budget framework of over ¥1 tn has already been secured. Regarding foundation models, development is progressing with Toyota’s LBM and SoftBank/Honda’s Noetra. Toyota plans to commence full-scale utilization of robots in its factories from around 2028, positioning the robotics business as a growth driver.

Exhibit 4: Japan is a leading country in robotics

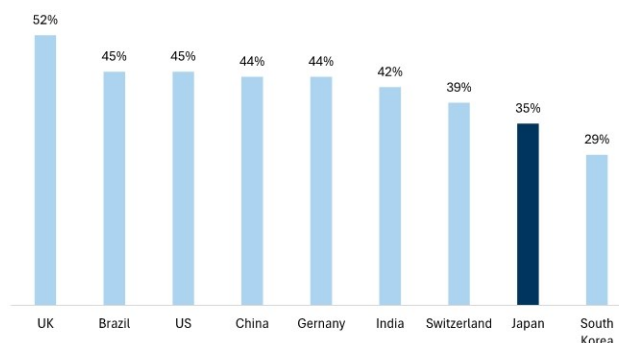
Number of operational robots per 10,000 workers



Source: World Robotics, Goldman Sachs Global Investment Research

Exhibit 5: Japan has a high level of receptiveness to robotics

The proportion of respondents who expressed concern about robots



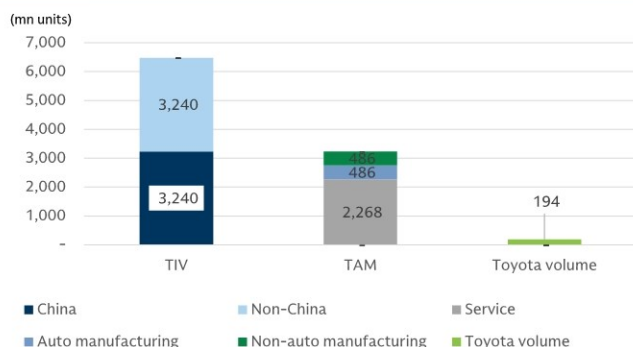
Source: Haxagon, Goldman Sachs Global Investment Research

Toyota’s humanoid market forecast

Despite being a late entrant, Japan possesses the social needs and foundational technologies. There is also a high possibility that the key players in Japan’s humanoid production will be centered around the automobile and auto parts industry, including Toyota, which has mass-produced industrial products by fully utilizing complex supply chains. The Toyota Group in particular possesses all four elements necessary to focus on the robotics business: Needs (securing a labor force), Urgency (addressing supply chain decline due to the shift to BEVs), Technology (hardware/software and mass production), and Commitment (management positioning robotics as a growth strategy, and financial strength). Assuming smooth technological breakthroughs and successful mass production ramp-ups, Toyota could potentially achieve a volume of approximately 190,000 to 540,000 humanoid robot units in 2035, capturing a global market share of approximately 3% to 8%. We estimate the incremental upside to our gross profit in 3/2027 to be 2% to 6%.

Exhibit 6: 2-6% upside opportunity for Toyota's gross profit for 2035

Toyota earnings simulation

Base case**Earnings impact (bn, JPY)**

Toyota Robotics sales	664
GM	30%
Gross profits	199
% of total (3/2027GSE)	1.7%

*Not assuming business development in China.

*Assuming 5%/10% market share for service/manufacturing robots

Blue-sky**Earnings impact (bn, JPY)**

Toyota Robotics sales	1,854
GM	40%
Gross profits	742
% of total (3/2027GSE)	6.2%

*Not assuming business development in China.

*Assuming 10%/25% market share for service/manufacturing robots

Source: Company data, Goldman Sachs Global Investment Research

Humanoid robotics will likely drive valuation multiple expansion

Our US and Korean research teams apply a P/E multiple of 150x for Tesla and 60x for Robotis to their respective robotics business earnings, factoring in growth potential for humanoid robotics valuation. For late-entrant Japanese automakers, the revenue contribution from humanoid robotics is projected to materialize in 2035 at the earliest. Discounting the P/E of 105x (average of Tesla and Robotis) using a WACC of 7%, the target P/E multiple for their robotics business comes to 57x. This suggests that if humanoid-related business eventually accounts for 5% of Toyota's consolidated revenue as mentioned earlier, Toyota's overall P/E multiple could potentially expand by over 20%. Needless to say, the key to unlocking this valuation upside lies in how concretely and effectively they can execute their robotics business plans.

Exhibit 7: Revenue contribution from humanoid robotics will drive valuation multiple expansion

P/E simulation

Valuation simulation	
Target P/E	105
Tesla	150
Robotis	60
WACC	7%
Discount	1.8
PV of target P/E	57

2035 GSE	Earnings contribution (% of total)				
P/E upside	0%	1%	5%	10%	20%
Toyota	0%	6%	23%	23%	38%
OEM (a)	0%	10%	38%	34%	51%
OEM (b)	0%	3%	11%	12%	22%
Target P/E	0%	1%	5%	10%	20%
Toyota	8.0	8.5	10.5	12.9	17.8
OEM (a)	5.0	5.5	7.6	10.2	15.4
OEM (b)	15.0	15.4	17.1	19.2	23.4

Source: Company data, Goldman Sachs Global Investment Research

Key risks to our view

Excessive expectations for on-site data/master craftsmanship: The accumulation of high-quality on-site data will likely play an important role in reproducing master craftsmanship. However, the foreseeable scope of applications for humanoids is relatively close to simple tasks, with a broad base. If efforts are focused on reproducing master craftsmanship, there is a risk of overlooking a massive potential market.

Reliance on overseas Foundation models: By quickly adopting open Foundation models, it is possible to speed up humanoid development. However, there are debates over whether they will remain open forever and whether the expansion of the digital deficit can be tolerated, presenting a risk that external reliance is not a sustainable business model.

Strong successful experiences in FA: The Japanese manufacturing industry has succeeded in the FA sector with relatively simple, repetitive tasks in mass production processes. However, the world of humanoids requires multipurpose and zero-shot responses by fully utilizing AI capabilities. There is a risk that successful experiences in FA will not necessarily be valued.

Exhibit 8: The group of companies supporting Japan’s humanoid robots
List of companies (not necessarily affiliated with T-HR3)



Humanoid robots
●Toyota
●Honda
●Mitsubishi Motors
●Softbank
○KyoHA
Foundation models
●Toyota
○Noetra
○Preferred Networks
●NEC
●Fujitsu
Dexterous hands
●Honda
Actuators
●Denso
●Aisin
●JTEKT
●Harmonic Drive
●NTN
●NSK
●Nidec
●Yaskawa
●Fanuc
●Mitsubishi Elec.
●MinebeaMitsumi
○Astemo
Sensors
●Murata Mfg.
●Denso
●Sony
Semiconductors
●Renesas
●Rohm
●Fuji Elec.
Batteries
●Panasonic
●TDK
○Toyota Battery
○Envision AESC
●Listed
○Non listed

The T-HR3 is Toyota’s third-generation humanoid robot. It stands 54 cm tall, weighs 75 kg, and features 32 axes of movement, ten fingers, and a remote-control system.

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 9: Japanese Stocks in focus

Humanoid beneficiaries

Position in value chain	Company	Ticker	GS Analyst	Rating	Mkt cap (JPYbn)	YTD Performance	Revenue (FY25, JPYbn)	P/E	P/B	Other business segments
Humanoid Robots	Toyota	7203.T	Kota Yuzawa	Buy	37,123	-12%	50,685	8.4	0.85	+Foundation models
	Honda	7267.T	Kota Yuzawa	Buy	6,632	2%	21,797	6.7	0.65	+Dexterous hands
	Mitsubishi Motors	7211.T	Kota Yuzawa	Sell	518	-3%	2,897	16.4	0.55	
	SoftBank Group	9984.T	-	NC	33,610	23%	7,799	58.0	1.78	
	KyoHA	Unlisted	-	-	-	-	-	-	-	
Actuators	Denso	6902.T	Kota Yuzawa	Neutral	4,853	-11%	7,540	10.1	0.83	+Sensors
	Aisin	7259.T	Kota Yuzawa	Buy	1,573	-27%	5,118	9.4	0.69	
	JTEKT	6473.T	Kota Yuzawa	Buy	658	22%	1,925	10.3	0.76	
	Harmonic Drive	6324.T	Yuichiro Isayama	Buy	539	67%	60	40.4	5.70	
	NTN	6472.T	-	NC	231	6%	826	12.1	0.86	
	NSK	6471.T	-	NC	572	16%	912	14.1	0.80	
	Nidec	6594.T	Daiki Takayama	Buy	3,004	26%	2,626	13.1	1.53	
	Yaskawa	6506.T	Yuichiro Isayama	Buy	1,247	4%	542	17.5	2.33	
	Fanuc	6954.T	Yuichiro Isayama	Sell	5,621	0%	858	30.6	2.94	
	Mitsubishi Elec.	6503.T	Ryo Harada	Buy	11,328	21%	5,895	18.6	2.24	
	MinebeaMitsumi	6479.T	Daiki Takayama	Buy	1,373	20%	1,664	12.5	1.26	+Bearings, Sensors
	Astemo	Unlisted	-	-	-	-	-	-	-	
Sensors	Murata Mfg.	6981.T	Daiki Takayama	Buy	14,149	121%	1,831	22.6	3.74	+MLCCs
	Sony	6758.T	Minami Munakata	Buy	23,581	-11%	12,480	16.8	2.48	
Semiconductors	Renesas	6723.T	Daiki Takayama	Buy	4,560	82%	1,655	8.3	1.09	
	Rohm	6963.T	Daiki Takayama	Buy	1,835	95%	481	21.0	2.12	
	Fuji Elec.	6504.T	Ryo Harada	Sell	2,044	19%	1,228	17.0	2.20	
Batteries	Panasonic	6752.T	Ryo Harada	Buy	10,098	112%	8,049	15.2	1.67	
	TDK	6762.T	Daiki Takayama	Buy	6,038	40%	2,505	18.8	2.19	+Sensors
	Toyota Battery	Unlisted	-	-	-	-	-	-	-	
	Envision AESC	Unlisted	-	-	-	-	-	-	-	
Foundation models	Fujitsu	6702.T	Chikai Tanaka	Buy	6,712	-11%	3,503	18.8	2.98	
	NEC	6701.T	Chikai Tanaka	Buy	6,515	-19%	3,583	18.0	2.36	
	Noetra	Unlisted	-	-	-	-	-	-	-	
	Preferred Networks	Unlisted	-	-	-	-	-	-	-	

P/E and P/B for coverage/non coverage are respectively based on FY27 GS estimates/BBG consensus

Source: Bloomberg, Company data, Goldman Sachs Global Investment Research

Japan's robotics industry as a "user"

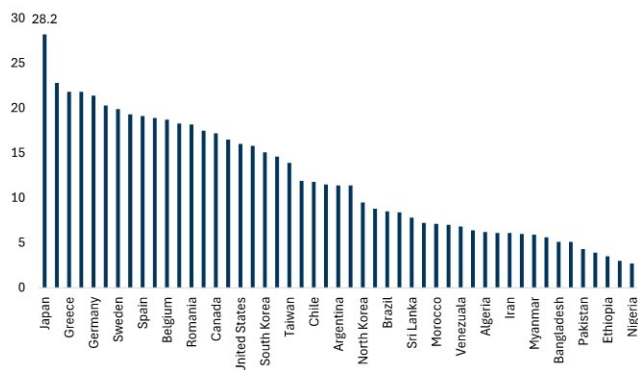
Severe labor shortage

Japan is one of the most rapidly aging countries in the world, with the highest population ratio of seniors aged 65 and over globally. At the same time, the manufacturing sector, including the automotive industry, accounts for around 20% of the economy's driving force, and labor shortages have become prominent in fields such as manufacturing, logistics, construction, and nursing care. Looking ahead to 2040, the labor force is expected to continue on a steady decline from its current level of around 65 million people. Social needs for humanoids are extremely high.

Labor issues are not a topic limited to Japan. In regions such as the US and Europe, there are many cases where a higher local parts procurement ratio is required (from an industrial protection perspective), making automobile and auto parts production necessary in regions with relatively high labor costs. Given tariff policies and the like, optimized production expansion, coupled with automobiles, will be an essential element in building a sustainable business structure.

Exhibit 10: Japan's Aging Population

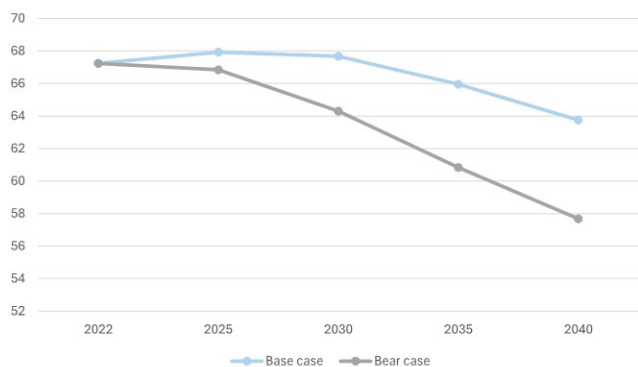
Mix of the population aged 65 and over



Source: World Bank

Exhibit 12: Decrease in workforce

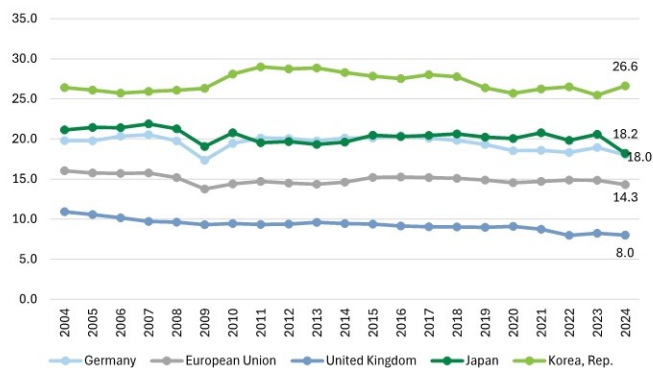
Estimated labor force figures



Source: METI, Goldman Sachs Global Investment Research

Exhibit 11: Manufacturing plays an important role in Japanese economy

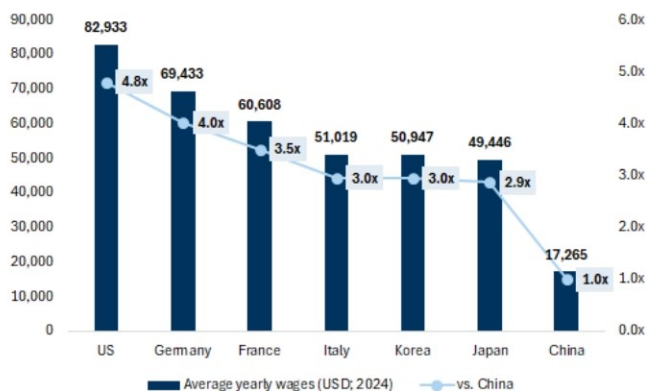
Share of manufacturing in GDP



Source: World Bank

Exhibit 13: Labor cost is high

Wage comparison (USD)



Source: Goldman Sachs Global Investment Research, OECD

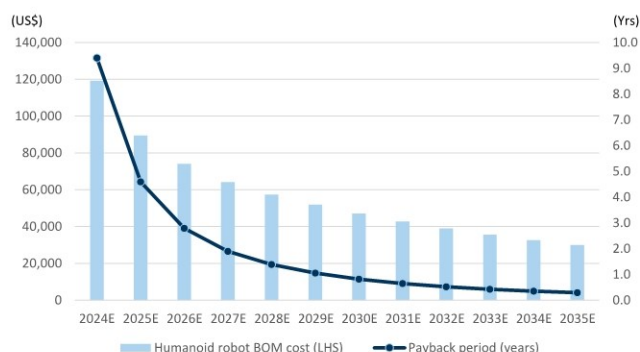
Declining robot prices to bring about a crossover point

Several humanoid robot manufacturers, such as UBTech, have suggested the possibility of humanoid costs dropping rapidly. As of 2026, we estimate the manufacturing cost (BOM: Bill of Materials) for industrial and research humanoid robots is around \$74,000 per unit, but similar to other industrial products, this cost will likely decline significantly through mass production. We forecast the price of humanoid robots to fall from \$42,000 in 2025 to \$21,000 in 2035.

As a result, we think we are fast approaching a “crossover point” where rising labor costs will intersect with the falling price of the robots themselves, shifting away from the current situation where robot implementation is not cost-effective. We analyze that the payback period for humanoid robots will fall to the equivalent of 2.8 years of labor costs in 2026, and to the equivalent of 1.9 years in 2027.

Exhibit 14: Cost for humanoid robots is expected to decrease

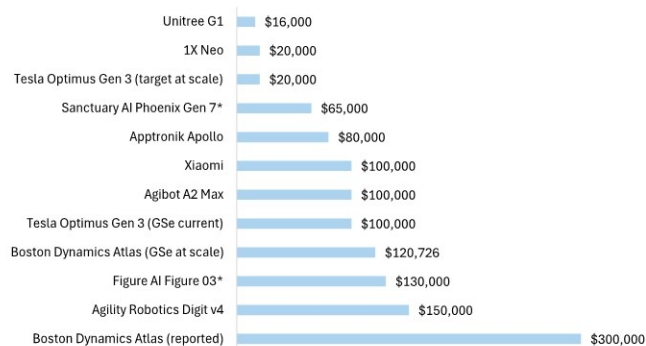
Cost and payback estimates



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 15: Wide price range

Estimated humanoid retail price



some prices quoted are expected prices at scale production, as of 2026

Source: Various media reports, Goldman Sachs Global Investment Research

Balance between price and performance is key

We think the societal implementation of humanoid robots, which will start with the automotive industry and ripple out to other industries, will not spread to homes and streets all at once. A scenario of gradual proliferation in accordance with safety, cost, and technical maturity is likely realistic. First, automobile factories and large-scale logistics hubs with a pressing need for automation will function as early adopters, and we view the data accumulated here and the cost reductions driven by mass production effects as serving as a catalyst for their spread to other industries.

Phase 1: Automobile factories / Large-scale logistics hubs (2026–2028)

- Closed environments where safety management is easy (inside fences or restricted areas).
- Mainly parts transportation, container loading/unloading, and simple picking tasks.

Phase 2: General manufacturing / Small-to-medium factories / General warehouses (2028–2032)

- Decline in unit prices due to mass production effects (assumed at \$20,000 to \$30,000 level per unit).

- Establishment of safety standards as collaborative robots, executing tasks in environments mixed with humans.

Phase 3: Services / Nursing care / Infrastructure / Homes (2032-)

- Full-scale rollout begins in non-manufacturing sectors where labor shortages are severe.
- Taking on autonomous operations in open environments where an unspecified large number of people are present.

Exhibit 16: Large workforce in the manufacturing, retail, and medical/welfare sectors

Workforce distribution in Japan

(mn ppl)	Management	Professional and technical occupations	Administrative	Sales	Service	Agriculture, Forestry, and Fisheries	Production process	Transportation and Machinery Operation	Mining and Construction	Transport, cleaning, packaging, etc.	Total
Agriculture, forestry, and fisheries	0.0	0.0	0.1	0.0	0.0	1.8	0.0	0.0	0.0	0.1	1.9
Mining and Construction	0.2	0.4	0.9	0.2	0.0	0.0	0.4	0.1	2.5	0.1	4.8
Manufacturing	0.2	1.1	2.0	0.5	0.0	0.0	6.1	0.1	0.0	0.5	10.4
Information and Communications	0.1	1.9	0.7	0.2	0.0	0.0	0.0	0.0	0.0	0.0	2.9
Transportation	0.1	0.0	0.8	0.1	0.0	0.0	0.1	1.3	0.0	1.0	3.4
Wholesale and retail	0.2	0.4	1.9	5.3	0.0	0.0	1.2	0.1	0.1	1.2	10.5
Finance, insurance, and real estate	0.1	0.1	1.3	1.0	0.3	0.0	0.0	0.0	0.0	0.1	2.9
Food & Beverage and Accommodation	0.0	0.0	0.1	0.2	3.4	0.0	0.0	0.0	0.0	0.2	4.1
Medical and Welfare	0.1	4.7	1.2	0.0	2.8	0.0	0.1	0.1	0.0	0.2	9.2
Education and Learning Support	0.0	2.5	0.6	0.0	0.2	0.0	0.0	0.0	0.0	0.1	3.5
Other services	0.1	1.6	2.3	0.3	0.2	0.0	0.6	0.2	0.0	1.3	6.8
Public and Composite Services	0.1	0.2	1.8	0.0	0.0	0.0	0.0	0.0	0.0	0.1	2.3
Total	1.2	13.2	14.2	8.1	8.5	1.9	8.6	2.1	2.8	4.9	65.6

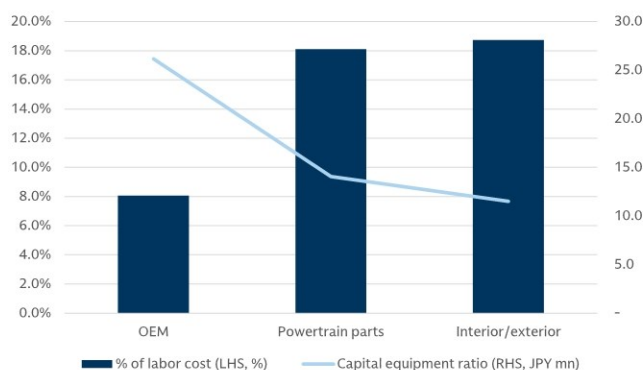
Source: METI, Goldman Sachs Global Investment Research

Focus on robot adoption for parts manufacturing in the automotive industry

Furthermore, in automobile production processes that are not currently robotized, the areas where humanoid robots can be applied remain at a mere 2% of the total (parts sorting and simple logistics/transportation tasks). To adapt humanoids to the remaining 98%, which consists of advanced processes such as “complex parts assembly,” “wiring and connector connection,” and “visual/abnormal noise inspections,” innovations in AI capabilities and robot hands are indispensable. In the automotive industry in particular, parts manufacturing is labor-intensive. While the transition to Phase 2 is highly likely to contribute significantly to alleviating labor shortages, curbing labor costs, and improving production quality, it will largely depend on the pace of technological innovation.

Exhibit 17: Parts manufacturing is labor-intensive in the automobile industry

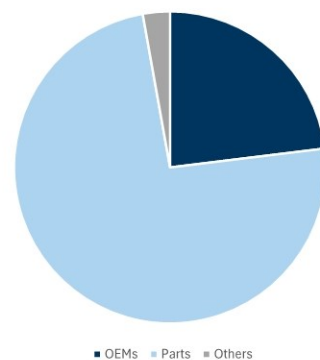
Labor cost mix comparison



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 18: Auto parts account for the majority of workforce in the auto industry

Distribution of workforce in the automotive industry



Source: METI, Goldman Sachs Global Investment Research

Humanoid Robots Offer Margin Upside for Manufacturing

The introduction of humanoid robots is poised to deliver a substantial margin upside for

the manufacturing sector. Needless to say, the labor cost-reduction rate will vary significantly depending on the price and performance of the robots, such as their continuous operating hours and multi-tasking capabilities. Taking the automotive and auto parts industry as an example, where the labor cost ratio typically ranges between 10% and 20%, we have set the baseline labor cost ratio at 15% for this simulation. The simulation was conducted with robot unit prices ranging from ¥3 mn to ¥9 mn. Assuming a robot adoption rate of 10% to 50%, we estimate 1-6% upside to operating profit margins.

Further upside opportunities include:

- **Continuous Operation:** While the operating days in this model are set at 250 days per year, robots can theoretically operate 365 days a year.
- **Performance Evolution:** The potential for robots to handle multi-tasking as their capabilities improve.
- **Productivity Gains:** Efficiency improvements driven by enhanced production quality and lower defect rates.

Exhibit 19: Returns Vary by Price and Performance

Estimation of Robot Cost-Reduction Effects (for illustrative purposes)

Cost reduction effects from Robot		
Robot Unit price	4,500,000	JPY
System integration costs	50%	
Annual maintenance costs	15%	
Power consumption	1	kW
Electricity rate	30	JPY/kWh
Average annual wage	8,000,000	JPY
Working days	250	Day
Working hours	8	Hours
Robot work capacity (relative to human)	25%	
Robot operating days	250	Day
Robot operating hours	6,000	Hours
Human equivalent working hours	1,500	Hours
Displaced full time equivalents	0.8	Person
Annual labor cost savings	6,000,000	JPY
Initial investment	6,750,000	JPY
Annual OPEX	855,000	JPY
Annual depreciation	900,000	JPY
Net annual savings	4,245,000	JPY
Payback period	1.6	Year

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 20: Operating Margin Expansion of 1% to 6%

Simulation of Labor Cost-Reduction Effects (for illustrative purposes)

Robot ASP (JPY)	Adoption rate	Cost reduction	Margin expansion
4,500,000	10%		1.2%
	25%	78%	2.9%
	50%		5.9%
6,000,000	10%		1.1%
	25%	72%	2.7%
	50%		5.4%
9,000,000	10%		0.9%
	25%	58%	2.2%
	50%		4.4%

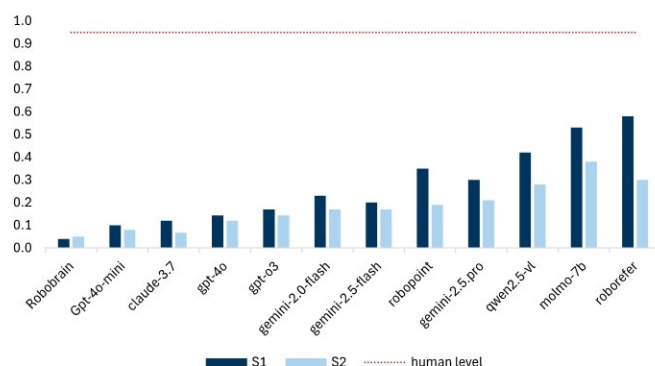
Source: Company data, Goldman Sachs Global Investment Research

Japan's robotics industry as a “creator”

Expanding the humanoid market requires the evolution of AI capabilities and the evolution of hardware (especially the Hands). When comparing object localization + actuation across multiple AI models to human capabilities, a significant gap still exists (though the difference is narrowing). Unlike conventional LLMs that are completed in digital spaces such as text and images, the humanoid brain, which repeats cognition, judgment, and action in real time in physical space, will require breakthroughs in “data quality and quantity” as well as “model lightening/optimization (distillation).”

Exhibit 21: Humanoid robots have not yet reached a level of performance that surpasses humans.

Comparison of proficiency levels for human capabilities (data as of Sep 2025)



S1 consists of relatively basic tasks (such as standing, walking, and reaching). S2 involves a combination of multiple skills.

Source: Xue, H., Ge, Y., Zeng, Y., Li, Z., Liu, M.-Y., Chen, Y., & Fan, J. (2025). Point-It-Out: Benchmarking embodied reasoning for vision language models in multi-stage visual grounding. arXiv.

Focus also on the development of robot hands

In addition, physical AI requires “Sim-to-Real,” which involves executing actions learned on a simulator in the real world without error, along with the minimization of latency. The human hand, in particular, consists of 27 bones and numerous muscles/nerves per hand, possessing an extremely delicate control of force. Current robot hands are capable of simple actions such as grasping and placing, but tasks such as “pinching a thin wire harness with fingertips and inserting it into a narrow gap with appropriate force” are difficult. “Next-generation robot hands,” equipped with over 20 degrees of freedom per hand and high-density tactile sensors on the fingertips, are drawing attention as the next breakthrough.

Areas where Japanese technology can be leveraged

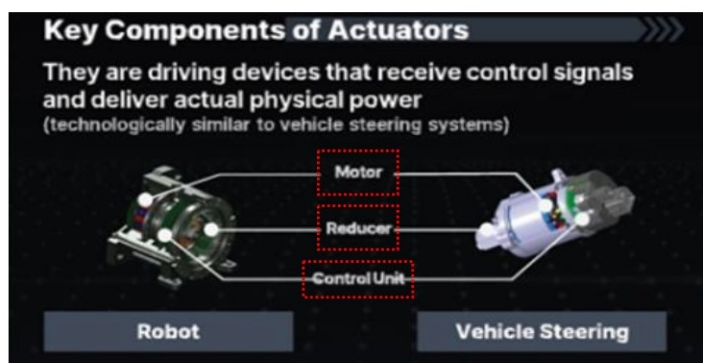
We analyze that the areas where Japan's hardware technology can be leveraged will be precision reduction gears, weight reduction, delicate robot hands, and quality control during mass production.

- Japanese companies boast a high market share globally in “precision reduction gears” that control the joints of humanoid robots. The performance of this core component dictates the robot's kinetic performance.

- In the unlikely event that a humanoid robot (assuming a height of 150-180cm and a weight of 50-80kg) falls over, there is a risk of causing immense damage to surrounding people and equipment. In addition, weight reduction is important because heavier robots consume correspondingly more power and have shorter operating times.
- Robot hands require the extremely high-density integration of microscopic actuators, reduction gears, tactile sensors, and wiring. There is also a possibility that the dexterity of a human hand cannot be achieved simply by combining modularized parts.
- Mass production know-how synchronized across the entire supply chain, represented by the Toyota Production System (TPS), also has the potential to lead to an advantage for Japanese companies in the phase when humanoid robots enter an era of mass production.

Exhibit 22: Compatibility with steering and actuators

Structure comparison



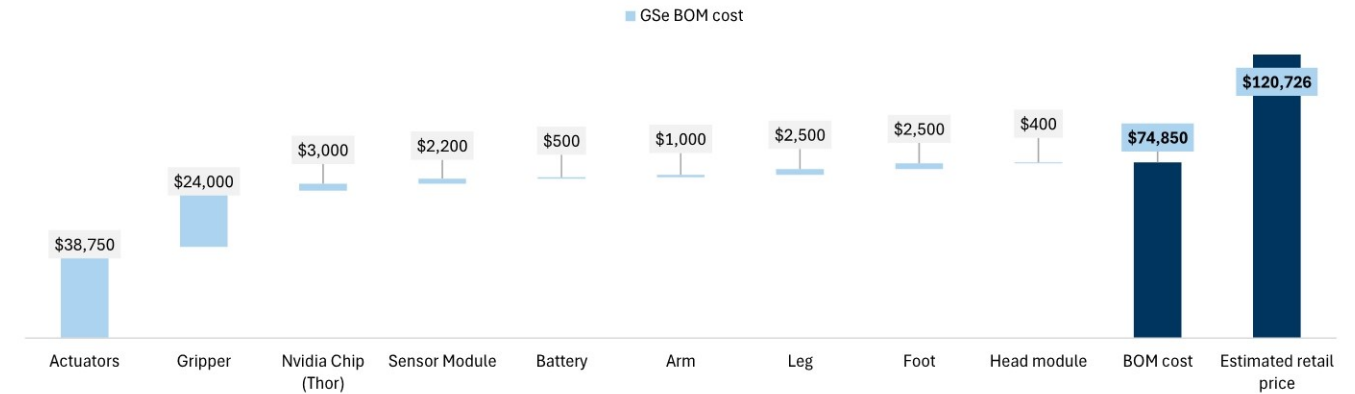
Source: Company data (Mando), Data compiled by Goldman Sachs Global Investment Research

An attractive market for the automotive industry

The automotive industry in particular possesses both needs and technologies that have a high affinity with humanoid robots. Generally, the shift to BEVs reduces the number of parts from around 30,000 to about 20,000. Suppliers that handle engine parts (pistons, crankshafts, cylinder blocks, etc.) and transmission gear machining are exploring the repurposing of their existing equipment due to the shift to BEVs. They will likely push forward with pivoting their businesses to the robotics industry by leveraging technologies in automobiles, centered around the “actuators” and “precision reduction gears” used in robot joints.

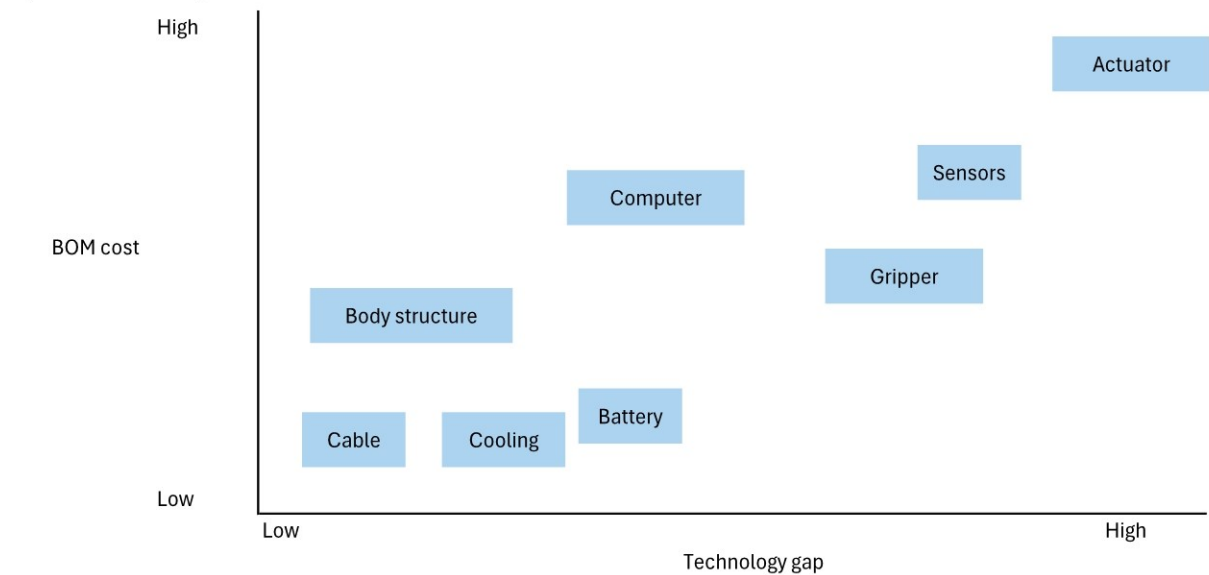
Analyzing the BOM cost of Boston Dynamics’ Atlas humanoid, actuators account for 52%, making them the highest value-added component. This is followed by grippers at 32%. NVIDIA semiconductors are 4%, other sensors are 2%, and batteries are 1%. While batteries account for 30-40% of BOM costs in BEVs, humanoids are characterized by a high ratio of mechanical hardware such as actuators and grippers. Furthermore, the component parts of actuators consist of motors, bearings, encoders, gearboxes, and drivers, which have a particularly high affinity with traditional auto parts.

Exhibit 23: Actuators and grippers offer high added value
BOM analysis based on Boston Dynmaics Atlas data



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 24: Distribution of value added
Key Humanoid Components



Source: METI, Goldman Sachs Global Investment Research

Unraveling Toyota’s strategy

The Toyota Group possesses all four elements necessary to focus on the robotics business: Needs (securing a labor force), Urgency (addressing supply chain decline due to the shift to BEVs), Technology (hardware/software and mass production), and Commitment (management positioning robotics as a growth strategy, and financial strength). President Kenta Kon is focused on achieving an ROE of 20%, and that focus means the company will emphasize the appropriate management of shareholders’ equity (including shareholder returns) as a matter of course, but the company’s strategy also signals a stronger message of embracing “discontinuous” growth. In addition to its existing business of manufacturing and selling vehicles, the company outlined the following as potential discontinuous business areas: 1) revenue diversification (Software Defined Vehicles, SDVs), 2) expansion of the robotics business by leveraging its manufacturing capabilities, and 3) accelerating the development of autonomous driving

from an AI-based approach as opposed to rule-based. We understand that the acceleration of robotics and AI development is a key focus of President Kon, who has gained experience in this regard from his time at Toyota subsidiary Woven.

(User)

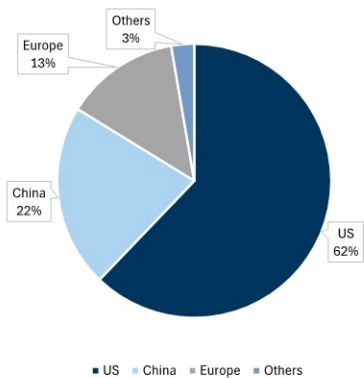
Toyota plans to begin introducing humanoid robots into its own factories in Japan by around 2028. We believe it will aim to roll them out horizontally to its global manufacturing bases thereafter. We note that concrete societal implementation has already begun. As of 2026, demonstration experiments for parts logistics/container handling using US-based Agility Robotics' humanoid robot "Digit" have started at Toyota's Woodstock plant in Canada (an assembly base for the RAV4).

(Creator)

The joint research partnership between TRI and Boston Dynamics announced in October 2024 is also noteworthy. This is an attempt to install the "Large Behavior Model (LBM)" developed by Toyota into Boston Dynamics' new humanoid, "Atlas." Toyota's Foundation Model, "LBM1 (Large Behavior Models 1.0)," serves as the foundation model for "Embodied AI," allowing the robot to integrate language, images, and its own senses to move autonomously in the physical world. We believe Toyota is currently the only Japanese automaker working on an in-house Foundation Model.

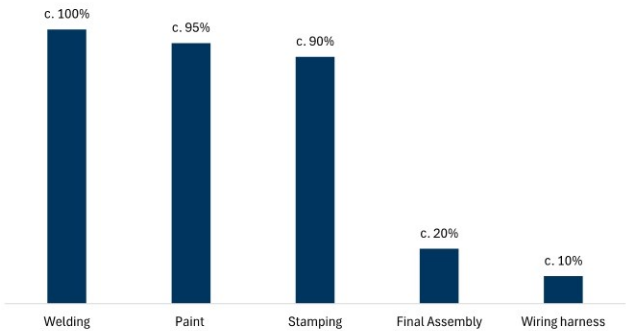
We conceptualize the architecture of Foundation Models in Physical AI through a distinct three-layered framework. **The World Model Layer:** This layer serves as the cognitive and predictive "brain" of physical reasoning, understanding fundamental physical laws and predicting future states. Key examples include NVIDIA's Cosmos and Toyota's Unified World Model (UWM). Major global technology firms, such as Google, Meta, and Tesla, are actively competing to establish dominance in this foundational layer. **The Digital Twin Layer:** This layer functions as the simulation, training, and validation environment (the "sandbox"). It is represented by platforms like NVIDIA's Omniverse and is primarily developed by factory automation (FA) giants such as Siemens, alongside various US and Chinese industrial software manufacturers. **The Industrial OS Layer:** This layer acts as the execution and control system (the "cerebellum"), translating high-level decisions into precise, real-time physical actions and motor control. This is equivalent to NVIDIA's Isaac platform. Under this framework, we categorize the Large Behavior Model (LBM) recently demonstrated by Toyota as a key technology belonging to this Industrial OS layer.

Exhibit 25: Low presence of Japanese manufacturers
of foundation models by region



Source: Humanoid.guide, Data compiled by Goldman Sachs Global Investment Research

Exhibit 26: Automation of the final assembly process has not progressed
Automation rate in automobile production



Source: Company data (Hyundai Motor), Goldman Sachs Global Investment Research

The Japanese government's ambitious robotics strategy

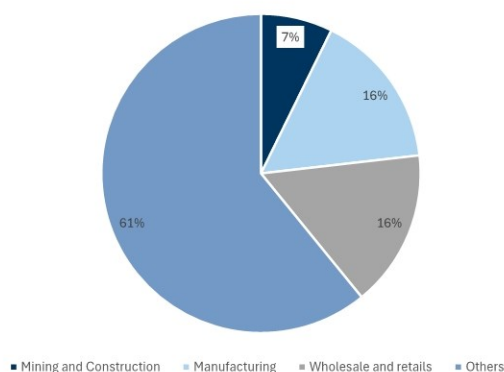
Setting ambitious targets

The Ministry of Economy, Trade and Industry expects the overall market for multi-purpose robots, including humanoids, to grow to around ¥60 tn globally by 2040, and aims for Japanese companies to capture a 30% share, or ¥20 tn, of this market. In the AI Robotics Strategy revised on June 30, 2026, it outlines (1) introducing approximately 10 mn AI robots domestically as a societal implementation target for 2040, and (2) introducing 350,000 units as a medium-term milestone for 2030. The five leading sectors will be “manufacturing,” “construction and civil engineering,” “architecture,” “retail,” and “security.” At the same time, the Japanese government has secured budget to support the development of the “brains” and “bodies” of robots that move autonomously in the real world.

The industries where the Japanese government aims to strategically introduce AI robots—such as construction, manufacturing, and retail—account for approximately 40% of the total workforce in Japan. Among these, tasks highly suited for robotization have been identified, including clerical work, sales, services, production, transportation, construction, and cleaning/packaging. Consequently, the immediate addressable market is expected to reach a scale equivalent to a workforce of 26 mn people. Under the assumption that it is difficult for a single robot to completely replace human labor, the target of introducing 10 mn AI robots by 2040 appears to be a realistic figure.

Exhibit 27: Initial Focus on 40% of the Workforce

Priority Sectors for Robot Adoption

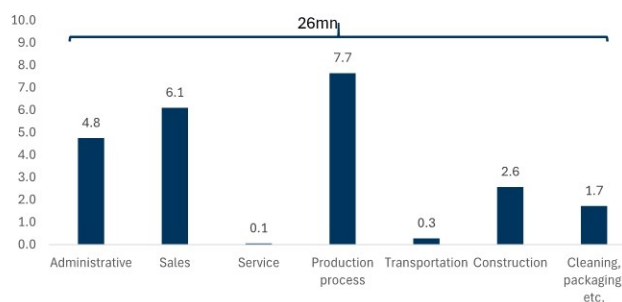


Source: METI, Goldman Sachs Global Investment Research

Exhibit 28: Scope of Tasks Relatively Easy to Robotize:

Approximately 26 mn People

Tasks Expected for Early Robotization



Source: METI, Goldman Sachs Global Investment Research

¥1 tn support budget scale for the development of domestic physical AI (multimodal foundation models)

Plans for a total of around ¥1 tn in support over five years (FY2026-FY2030).

- ¥387.3 bn has been allocated in the budget proposal for FY2026, the first year.
- Developing entities: Selected a consortium consisting of a new company, “Noetra,” funded by SoftBank, NEC, Sony Group, Honda Motor, and others, as well as the National Institute of Advanced Industrial Science and Technology (AIST).
- Objective: To develop a “multimodal foundation model” capable of integrated

processing of language, images, and video as well as robot sensor information and tactile data, and to build a domestic AI foundation leveraging the strengths of on-site data from Japan's manufacturing, nursing care, and disaster response sectors.

Individual support for next-generation robot hardware (robot hands, etc.)

Providing approximately ¥10 bn in support for Honda's robot hand development. This involves the development of an "advanced robot hand equipped with tactile sensors," proceeding jointly with RIKEN based on the technology of the bipedal robot "ASIMO", and has been selected as part of the physical AI project and will receive support of around ¥10 bn.

Honda announced a new multi-fingered robot hand at the Humanoid Summit Tokyo 2026. It named its drive system "Willow Drive". Existing robot hands present a choice between two options: a direct motor connection system (dexterous but weak) or a wire drive system (strong but with inferior precision), but this system drives the fingers with a proprietary transmission mechanism while placing the motor in the forearm (details undisclosed). Its strengths include a maximum fingertip force of 12kg, twice that of an adult male, 20 degrees of freedom (18 motors), and an operational durability of 8 mn cycles.

Exhibit 29: Japan is setting ambitious goals.

Comparison of policies across countries

	Japan	US	China
Policy / Initiatives	- Integrated design of public-private collaboration and supply-demand alignment - Resolution of social issues with the strengthening of industrial competitiveness against the backdrop of a severe labor shortage - Leverage government demand, such as for defense and disaster response, as initial demand.	- National Security, AI Supremacy, and Federal Procurement - Protection of the domestic industry through measures such as the U.S. Security Robotics Act (ASRA). - Prioritizing U.S.-made products in federal procurement via executive order; formulation of a national strategy is currently underway.	- State-led, massive investment, vertical integration - Robotics and AI (Embodied AI) have been positioned at the core of the 15th Five-Year Plan (2026–2030). - Strong backing from government-affiliated funds and others on the scale of one trillion yuan.
Industry cultivation	- Component strengths + standardization - Leveraging strengths in areas such as motors and reduction gears to transit from tight coupling to loose coupling (standardization).	- Extending the overwhelming superiority of software and AI models—such as LLMs and VLAs—into the physical world. - VC funding for startups	- Massive-scale production, cost reduction, and rapid validation. - Mass production of ultra-low-cost humanoids by companies such as Unitree and UBTECH.
Market outlook	20 tn yen (30% market share) by 2040 - Against the backdrop of a severe labor shortage, the aim is to implement these solutions first in essential sectors such as logistics, construction, and nursing care.	- High value-added, defense, and advanced industries - Reshoring (bringing manufacturing back to the domestic market) and addressing labor shortages. - Focus on the defense, space, and advanced manufacturing sectors.	- The World's Largest Robot Market. Expanding exports - Accounting for more than 54% of global industrial robot installations. - Producing 90% of the world's humanoids by 2025. Sweeping the global market with low-cost products.

Source: METI, Goldman Sachs Global Investment Research

Exhibit 30: Humanoid robot manufacturers

Company name	Country	Founded	Humanoid robot name	Partnership
Agility	US	2015	Digit	Amazon, Toyota
Apptronik	US	2016	Apollo	Mercedes
Figure	US	2022	Figure01/02	BMW
Tesla	US	2003	Optimus	Tesla
Boston Dynamics	US	1992	Atlas	Hyundai
Sanctuary AI	Canada	2018	Phoenix	Canadian Tire
Xpeng	China	2014	Iron	Xpeng
UBTech	China	2012	Walker S, S1, X ,C	BYD, Zeekr, Foxconn
Unitree	China	2016	H1, G1, R1	BYD, Honda

Source: Company data, METI, Compiled by Goldman Sachs Global Investment Research

Potential stock beneficiaries

Toyota (7203.T)

The Toyota Group possesses all four elements necessary to focus on the robotics business: Needs (securing a labor force), Urgency (addressing supply chain decline due to the shift to BEVs), Technology (hardware/software and mass production), and Commitment (management positioning robotics as a growth strategy, and financial strength). Assuming smooth technological breakthroughs and successful mass production ramp-ups, Toyota could potentially achieve a volume of approximately 190,000 to 540,000 humanoid robot units in 2035, capturing a global market share of approximately 3% to 8%. We estimate the incremental upside to our 3/2027 gross profit to be 2-6%. We believe that executing their robotics business plans with greater specificity going forward could drive an upward revision in both earnings and valuation multiples.

Honda (7267.T)

We believe that Honda possesses deep interest and domain expertise in robotics development, given its history of pioneering the bipedal robot ASIMO long before Physical AI began driving transformative changes in the industry. Recently, Honda has focused heavily on the development of robotic hands driven by its proprietary “Willow Drive” mechanism, accurately identifying key bottlenecks (or, put differently, the primary source of value creation) in humanoid robotics and allocating its resources accordingly. As previously noted, Honda is also actively utilizing Japanese government support for robotic hand development. According to Nikkei reports, Honda intends to conduct field validation tests of its multi-fingered hand within its own manufacturing facilities, aiming for practical commercial deployment from 2030 onward. However, because current disclosures regarding the future volume scale and cost structure of Willow Drive remain limited, these developments do not yet warrant an upward revision to the company’s earnings forecasts or valuation. We await further information disclosures in the future.

MMC (7211.T)

In July 2026, the company announced a joint development and mass-production partnership for humanoid robots with Highlanders, aiming to establish a production capacity of 1,000 units per month by the second half of 2027. Under this plan, Mitsubishi Motors will handle the manufacturing of the humanoid robots, while Highlanders will primarily lead sales. The target market for this monthly output of 1,000 units will also include external customers beyond Mitsubishi Motors’ own production lines. While the automaker aims to utilize domestic suppliers as much as possible, management recognizes that building a robust supply chain—particularly for actuators and motors—remains a major challenge. The global race to develop humanoid robots is accelerating, led primarily by US and Chinese manufacturers, with rapid advancements in both AI capabilities and degrees of freedom (DOF, particularly in robotic hands). Leading up to 2027, we intend to closely monitor the competitiveness and production capabilities of the mass-produced Mitsubishi Motors x Highlanders humanoid robot. Regarding financial impact, management has indicated that this initiative is currently an investment for the future, with no near-term revenue contribution expected.

JTEKT (6473.T)

In the short term, we expect the company's earnings to be driven by Toyota's robust vehicle production increases and the positive effects of structural reforms in North America and Europe. In the medium term, sales of steer-by-wire systems—which the company commercialized ahead of its competitors—are expected to expand alongside the adoption of autonomous driving. Furthermore, looking toward 2035, there is significant growth potential in actuator products, which serve as the critical core for humanoid robot movements. Indeed, South Korean steering competitor HL Mando and major European bearing manufacturer Schaeffler have already commercialized actuators for humanoid robots by leveraging their existing proprietary technologies. JTEKT, which operates across three core business segments—steering, bearings, and factory automation (FA) systems—has also commenced development of future products targeted at humanoid robotics. Although there is no concrete timeline for commercialization yet, we surmise that Toyota's humanoid robot initiative will be an all-hands-on-deck project leveraging the collective strength of its group companies, in which JTEKT is poised to play a major role. In our bull case, this business has the potential to generate tens of billions of yen in operating profit by 2035.

Aisin (7259.T)

Aisin handles a broad product lineup including transmissions (manual transmissions, automatic transmissions, and eAxles), braking systems, and body components, with particularly strong technological expertise in metal processing (aluminum). In addition, Aisin mass-produces automotive rear-wheel steering systems (whereas front-wheel steering systems are supplied by JTEKT), and we believe that the company possesses extensive mass-production experience with highly durable, compact actuators. While Aisin's management states that entering the robotics domain—including humanoid robotics—remains in the evaluation phase, they have indicated that the technological capabilities cultivated through automatic transmissions and braking systems share strong synergies and affinity with actuators used in humanoid robots. Should momentum accelerate toward commercializing an actuator business from 2030 onward in alignment with Toyota's group-wide robotics strategy, it could serve as a key catalyst to uplift Aisin's financial performance and equity valuation.

Harmonic Drive Systems (6324.T)

Harmonic Drive Systems has stepped up its humanoid robotics initiatives since FY3/24, subsequently beginning to disclose AI robot-related orders and sales, and is now recognized in the Japanese equity market as the flagship humanoid play. From the early days—when it was uncertain whether precision reducers would be adopted across all humanoid joints—through the subsequent wave of technological innovation and the expansion of the humanoid supply chain in China — the company has consistently focused on how to sustain the value-added of its products at a high level. These efforts have borne fruit: while humanoid-related sales currently account for only around 3% of the total, they have at times represented 5–10% of consolidated orders on a quarterly basis.

Humanoid-related sales came in at ¥1.9bn in FY3/26, with FY3/27 guidance at ¥2.3bn, serving a customer base of roughly 15 companies centered on US humanoid startups. While revenue recognition has been relatively slow, 1Q consolidated orders got off to a strong start at ¥1.25bn, and management has expressed confidence in its outlook that order growth and an acceleration in sales growth—driven by major customers scaling up

production from CY27 onward—can be expected from 2H FY3/27. The company's FY3/29 humanoid sales target stands at ¥7.1bn, and we see scope for upside to this figure. We also note that, with the current customer portfolio consisting mainly of customers that recognize the value-added of the company's precision reducers, management consistently points out that this value-added remains reasonably well protected. Accordingly, we expect AI robots to contribute as one of the drivers of earnings growth from FY3/28 onward.

In terms of involvement in Japan's humanoid supply chain, the key area of interest is likely to be the development of actuators and the company's contribution thereto. While partner names have not been disclosed, the company is actively engaged in developing actuators primarily for humanoid applications, not only with overseas companies but also with Japanese firms. The rationale for pursuing partnerships rather than going alone lies in scale: once the humanoid market expands to the size implied by an eventual mass-adoption phase, conventional production methods—premised on the scale of existing markets such as industrial robots and semiconductor production equipment—would be insufficient to keep pace. Of the humanoid sales figures noted above, Japan-related sales were ¥0.2bn in FY3/26, with plans calling for ¥0.3bn in FY3/27 and ¥3.3bn in FY3/29. Going forward, we will be watching whether the company can capture the benefits of the expansion of Japan's humanoid supply chain.

NEC (6701.T)/Fujitsu(6702.T)

Regarding initiatives in physical AI, the developments at Fujitsu and NEC warrant close attention.

Fujitsu has commenced business discussions with FANUC, Yaskawa Electric, and Kawasaki Heavy Industries to explore the societal implementation of physical AI incorporating technology from US-based NVIDIA. The company plans to deliver a core system by combining AI servers (featuring Fujitsu CPUs and NVIDIA GPUs)—which serve as the computing infrastructure to run AI models—with software components such as AI platforms, AI agents, physical OS, and collaborative control platforms. Specifically, through these AI servers, the collaborative control platform will transmit AI decisions to real-world robots and equipment, linking multiple devices (such as robots and factory equipment) and systems via networks to achieve collaborative operations and autonomy. Fujitsu intends to co-develop this collaborative control platform with its partners, with the goal of standardizing it and making it open-source. Considering the timing of the start of AI server manufacturing (Q1 2027), the physical AI business is expected to make a full-scale contribution to earnings in the fiscal year ending March 2028 (FY3/28). Anticipated applications include factory robots (optimizing overall production planning and enabling autonomous on-site adaptation), retail and logistics robots (automating transport operations based on logistics plans), and healthcare robots (autonomously executing instructions from hospital operational systems). As the core system controlling these robots, this business is expected to drive earnings growth.

Meanwhile, NEC plans to engage in the physical AI business as a core member of “Noetra,” a corporate consortium aiming to develop a domestic multimodal foundation model that will serve as the foundation for AI-enabled robots and physical AI. Noetra commenced full-scale operations in July 2026. In April 2027, it plans to begin constructing an AI computing infrastructure equipped with approximately 27,500 of NVIDIA's latest “NVIDIA Rubin GPUs,” with operations scheduled to start in June 2028.

NEC's role is to provide technologies such as video analytics, facial recognition, and AI inference, as well as to implement this common foundation model across various industries and enterprises. Consequently, we expect NEC to benefit through its system integration (SI) business.

MinebeaMitsumi (6479.T)

MinebeaMitsumi positions humanoid robots as one of its "Five Growth Areas," and aims to capture market opportunities through the integration of its wide-ranging technologies and product lineup. The company targets ¥100 bn in humanoid-related sales as a base case for FY3/35 (¥200 bn in an upside case), with a breakdown of ¥40 bn from bearings, ¥30 bn from motors, ¥20 bn from analog semiconductors, and ¥10 bn from sensors and other components. The plan highlights the potential match between the company's broad product portfolio and the component needs of humanoid robots.

The number of bearings installed per humanoid can be more than double that of a luxury car in some cases (a single hand can have 85 bearings), illustrating the high component density in MinebeaMitsumi's business domains. We believe that the company is winning orders for sensors and bearings from customers including major US players, and that inquiries for motors and various other components are also becoming more active. For robot hands in particular, the company aims to differentiate itself through the integration of its core technologies, and has commented that it is currently in joint development with three companies.

Renesas Electronics (6723.T)

Renesas views humanoid robots as an aggregation of four domains ("brain & motion," "sensing," "actuation & motor control," "power management"), offering products that correspond to each domain. Currently, the company's SAM covers about 30% of a humanoid's BOM. Against a backdrop of increasing robot complexity, such as the decentralization of intelligence and a rise in the number of joints and sensing needs, Renesas has outlined a plan to expand its coverage to 70% of the BOM in the long term.

Renesas has a distinct advantage in its ability to leverage proven technologies from the automotive sector. In addition, beyond its broad hardware portfolio, the company's competitive edge lies in its ability to support their customers' development through the "Renesas 365" platform, which facilitates design optimization in a virtual environment. As many players enter the humanoid market (a market with remarkable technological evolution characterized by high-mix, low-volume production and rapid change), we believe the value of "Renesas 365" (which enables efficient design, improvement, and lifetime management on a cloud platform) could be increasingly recognized. By delivering solutions that integrate hardware and software, the company aims to capture share in this expanding market. Currently, in addition to its strain gauges being adopted in the hands of China-based Inspire-Robots, Renesas is actively engaged with numerous customers globally.

Price Target Risks and Methodology - Toyota Motor

We are Buy rated on Toyota. Our 12-month target price of ¥3,900 is based on P/B-ROE correlation using our FY3/28E estimates. Key downside risks include yen appreciation, delays in EV development, battery sourcing issues amid rapid growth in the EV market, disappointment in the company's shareholder return policy, and low capacity utilization

in the China business.

Price Target Risks and Methodology - Honda Motor

We are Buy-rated on Honda. Our 12-month target price of ¥2,000 is based on a P/B-ROE correlation and our FY3/28E estimates. Downside risks include yen appreciation, a sharp rise in raw material prices, weaker-than-expected sales of new vehicles, inefficiency in multiple EV projects, and margin deterioration in the China business.

Price Target Risks and Methodology - Mitsubishi Motors

Our 12-month target price of ¥330 is based 85% on a fundamental value of ¥270, derived from the FY3/28E P/B-ROE correlation, and 15% on a theoretical M&A value of ¥660, which we base on the average P/B of 1.0X since Nissan made its capital investment in MMC (based on FY3/26E BPS). Key upside risks include (1) stronger sales trends in ASEAN markets, (2) an improvement in the US business through a deepening of alliances, and (3) yen depreciation.

Price Target Risks and Methodology - NEC

Our 12-month target price of ¥5,190 is based on our FY3/28E EPS of ¥243 and a target P/E of just over 21X. Our target P/E is derived from (1) a P/E of c.21-22X for the IT services business, which is a 20% premium to the average of 18.0X for the IT services sector, and (2) a P/E of c.20X for the hardware business, which is the sector average for major electronics companies. We then take a weighted average of these two multiples, based on the company's sales weightings, with a 90% weighting for IT services and a 10% weighting for hardware. We apply a premium to the average for the IT services sector because we look for profit growth above the sector average over the medium term, driven by growing demand for backbone systems and margin improvement from tackling low-margin projects, as well as structural expansion in defense business. Key risks include lower corporate IT investment appetite, weak telecom carrier capex, a shift to overseas products, and the emergence of unprofitable projects.

Price Target Risks and Methodology - Aisin

Our 12-month target price of ¥3,000 is based on a P/B-ROE correlation and our FY3/28 estimates. We are Buy-rated. Key risks include increase in aluminum prices, prolonged Middle East impacts, and price negotiations with major customers.

Price Target Risks and Methodology - Harmonic Drive Systems (6324.T)

Our 12-month target price of ¥11,000 is based on FY3/30E EV/EBITDA, applying the sector-average multiple of 10X and a 180% sector-relative premium, the same level observed in the previous peak cycle, with the resulting theoretical value discounted back to the midpoint of FY3/27E and FY3/28E at a capital cost of 10%.

Downside risks to our target price include (1) weaker investment appetite from manufacturers/EMS of tech products, and a decline in appetite for mass production at equipment manufacturers, primarily of industrial robots and SPE, and (2) greater commitment to the auto components business.

Price Target Risks and Methodology - MinebeaMitsumi

We are Buy-rated. Our 12-month price target is ¥5,500, based on FY3/29E EV/GCI vs. CROCI/WACC, applying the sector average EV/DACF multiple of 10X. Our target price implies FY3/28E P/E of 20X. Key risks include sharper-than-expected rises in input costs, a decline in auto production, and forex swings.

Price Target Risks and Methodology - Fujitsu

Our 12-month target price of ¥4,330 is based on our FY3/28 EPS estimate of ¥206 and a target P/E of 21X. Our target P/E is derived from (1) a P/E of c.21-22X for the IT services business (20% premium to the sector-average P/E of 18.0X), and (2) a P/E of 20.0X for the hardware business (average for the hardware sector). We then take a weighted average of these two multiples, based on the company's profit weightings, with an 80% weighting for IT services and a 20% weighting for hardware. The reasons we apply a premium to the sector average P/E are (1) a medium-term profit growth rate that significantly exceeds the sector average, (2) an aggressive shareholder return policy, (3) the company's favorable positioning to benefit from generative AI, including in generative AI-powered system development and AI solutions, and (4) growth stories not seen at other SIs, such as the company's defense business. Key risks include deterioration in IT services/hardware demand due to worsening business sentiment, weaker telecom carrier capex, and the emergence of unprofitable projects.

Price Target Risks and Methodology - Renesas Electronics

We are Buy rated on Renesas. Our 12-month target price of ¥5,500 is based on FY12/28E EV/GCI vs. CROCI/WACC, applying an EV/DACF multiple of 13X (30% premium to the sector average, implying a FY12/27E target P/E of 16X). Key risks include a slowdown or delayed recovery in consumer electronics-related demand, stagnant automobile production, protracted reduction in semiconductor inventories, delayed recovery for industrial equipment applications, and yen appreciation.

Price Target Risks and Methodology - JTEKT

We are Buy-rated with our 12-month target price of ¥2,700. This is derived from a weighted average of fundamental value (¥2,500; 85% weight; based on a 5x EV/EBITDA multiple) and M&A value (¥3,500; 15% weight; based on a peak P/B ratio of 1.5x since the Global Financial Crisis). Key risks include a decline in Toyota's production volume, yen appreciation, and further delays in Toyota's humanoid robot development.

Disclosure Appendix

Reg AC

We, Kota Yuzawa, Do Hyoung Kim, Mark Delaney, CFA, Christian Frenes, Chikai Tanaka, CFA, Daiki Takayama, Yuichiro Isayama, Ken Kawamoto and Joshua Kim, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Disclosures

Rating and pricing information

Aisin (Buy, ¥2,232), Denso (Neutral, ¥1,911), Fanuc (Sell, ¥6,029), Fuji Electric Co. (Sell, ¥14,210), Fujitsu (Buy, ¥3,763), Harmonic Drive Systems (Buy, ¥5,810), Honda Motor (Buy, ¥1,662), JTEKT Corp. (Buy, ¥2,050), MinebeaMitsumi Inc. (Buy, ¥3,404), Mitsubishi Electric (Buy, ¥5,586), Mitsubishi Motors (Sell, ¥375), Murata Mfg. (Buy, ¥7,273), NEC (Buy, ¥4,737), Nidec (Buy, ¥2,614), Panasonic Holdings (Buy, ¥4,362), Renesas Electronics (Buy, ¥3,375), Rohm (Buy, ¥4,840), Sony Group (Buy, ¥3,869), TDK (Buy, ¥3,075), Toyota Motor (Buy, ¥3,081) and Yaskawa Electric (Buy, ¥4,878)

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Company-specific regulatory disclosures

Compendium report: please see disclosures at <https://www.gs.com/research/hedge.html>. Disclosures applicable to the companies included in this compendium can be found in the latest relevant published research

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
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Price target and rating history chart(s)

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Target price history table(s)

Fujitsu (6702.T)

Date of report	Target price (¥)	Closing price (¥)
21-Aug-26	4,380	3,646
22-May-26	4,330	3,307
24-Feb-26	4,500	3,313
21-Nov-25	5,000	4,129
25-Aug-25	4,300	3,571
02-Jun-25	4,030	3,461
28-Feb-25	3,570	2,879
22-Nov-24	3,230	2,757
28-Aug-24	3,240	2,590
22-May-24	2,740	2,322
22-Feb-24	27,200	2,271
22-Nov-23	23,700	2,058

NEC (6701.T)

Date of report	Target price (¥)	Closing price (¥)
21-Aug-26	6,000	4,656
22-May-26	5,190	4,105
25-Feb-26	5,200	3,760
21-Nov-25	6,500	6,074
25-Aug-25	5,250	4,582
02-Jun-25	4,160	3,809
28-Feb-25	17,300	2,893
22-Nov-24	15,300	2,453
28-Aug-24	14,600	2,444
22-May-24	12,300	2,240
22-Feb-24	11,500	1,976
22-Nov-23	8,900	1,617

Renesas Electronics (6723.T)

Date of report	Target price (¥)	Closing price (¥)
31-Jul-26	5,500	3,442
08-Jun-26	5,400	4,139
24-Apr-26	3,800	3,141
23-Mar-26	3,600	2,276
05-Feb-26	3,300	2,767
12-Jan-26	2,900	2,380
24-Apr-25	2,500	1,670
31-Mar-25	2,900	1,988
06-Feb-25	2,700	2,306
31-Oct-24	2,600	2,110
01-Oct-24	2,900	2,161
25-Jul-24	3,400	2,584
02-Jul-24	3,800	3,014
03-Apr-24	3,400	2,650
08-Feb-24	3,200	2,542
27-Nov-23	3,300	2,563
26-Oct-23	2,900	2,106
04-Oct-23	3,000	2,148

Aisin (7259.T)

Date of report	Target price (¥)	Closing price (¥)
31-Jul-26	3,000	2,224
29-Apr-26	3,200	2,305
03-Feb-26	3,100	2,797
14-Dec-25	3,000	3,067
31-Oct-25	2,250	2,775
31-Jul-25	2,000	2,099
06-Jun-25	1,600	1,747
15-Apr-25	1,400	1,493
04-Mar-25	1,500	1,748
31-Oct-24	1,450	1,603
17-Sep-24	4,400	1,554
26-Apr-24	4,600	1,879
31-Oct-23	4,100	1,734

MinebeaMitsumi Inc. (6479.T)

Date of report	Target price (¥)	Closing price (¥)
08-Jun-26	5,500	4,546
12-May-26	4,000	3,566
23-Mar-26	3,800	2,541
12-Jan-26	3,700	2,997
06-Nov-25	3,400	3,114
02-Oct-25	3,300	2,764

Mitsubishi Motors (7211.T)

Date of report	Target price (¥)	Closing price (¥)
03-Aug-26	330	358
29-May-26	360	372
20-Apr-26	330	336
13-Jan-26	370	380
07-Jan-26	350	372
27-Aug-25	420	402

MinebeaMitsumi Inc. (6479.T)

Date of report	Target price (¥)	Closing price (¥)
05-Aug-25	3,000	2,395
09-May-25	2,900	2,177
31-Mar-25	3,000	2,174
05-Feb-25	3,300	2,447
06-Nov-24	3,400	2,858
01-Oct-24	3,500	2,941
02-Jul-24	3,700	3,552
03-Apr-24	3,400	2,907
02-Feb-24	3,200	3,101
06-Dec-23	3,100	2,831
02-Nov-23	2,800	2,504
04-Oct-23	2,700	2,398

Mitsubishi Motors (7211.T)

Date of report	Target price (¥)	Closing price (¥)
23-Jul-25	450	440
10-Apr-25	440	387
03-Feb-25	500	446
30-Oct-24	520	421
17-Sep-24	500	372
08-May-24	530	474
04-Apr-24	550	487
01-Feb-24	530	473
13-Dec-23	550	453
28-Sep-23	600	670

Harmonic Drive Systems (6324.T)

Date of report	Target price (¥)	Closing price (¥)
29-Jun-26	11,000	7,120
13-May-26	10,000	6,970
09-Mar-26	5,700	3,805
14-Jan-26	5,400	3,835
13-Oct-25	5,000	3,300
14-May-25	4,300	3,835
11-Apr-25	4,500	2,913
07-Feb-25	5,800	4,535
16-Jan-25	5,200	3,930
12-Nov-24	5,000	2,649
07-Oct-24	5,700	3,315
16-Aug-24	6,000	3,955
12-Apr-24	5,500	4,045
08-Jan-24	5,200	3,865
13-Sep-23	5,100	3,365

Honda Motor (7267.T)

Date of report	Target price (¥)	Closing price (¥)
05-Aug-26	2,000	1,636
02-Jul-26	1,850	1,527
20-Apr-26	1,800	1,341
10-Feb-26	1,900	1,670
13-Jan-26	2,000	1,604
05-Jan-26	1,850	1,558
09-Nov-25	1,950	1,585
08-Sep-25	2,000	1,694
10-Apr-25	1,750	1,349
21-Feb-25	2,000	1,380
06-Nov-24	2,100	1,418
17-Sep-24	2,200	1,457
07-Aug-24	2,300	1,423
03-Jul-24	2,400	1,739
10-May-24	2,300	1,737
04-Apr-24	2,200	1,849
08-Feb-24	2,100	1,734
11-Jan-24	2,000	1,569
28-Sep-23	2,100	1,731
18-Sep-23	6,000	1,750

Toyota Motor (7203.T)

Date of report	Target price (¥)	Closing price (¥)
20-Apr-26	3,900	3,390
13-Jan-26	4,000	3,641
05-Nov-25	3,800	3,040
08-Sep-25	3,500	2,970
23-Jul-25	3,200	2,855
08-May-25	3,100	2,672
10-Apr-25	3,300	2,543
17-Sep-24	3,800	2,406
01-Aug-24	4,000	2,699
03-Jul-24	4,500	3,325
08-May-24	4,300	3,579
27-Mar-24	4,400	3,853
06-Feb-24	3,600	3,135
01-Nov-23	3,300	2,712
28-Sep-23	3,200	2,706
21-Sep-23	3,100	2,786

JTEKT Corp. (6473.T)

Date of report	Target price (¥)	Closing price (¥)
30-Aug-26	2,700	2,035

Price targets shown in table(s) are unadjusted for corporate actions.

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